NEWTON SCHOOL

PRE-CALCULUS

Rational Functions: Asymptotes, Holes & Graphing

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Standard: CCSS.MATH.CONTENT.HSF-IF.C.7d — Graph rational functions, identifying zeros, asymptotes (vertical, horizontal, slant), and holes. Perform polynomial long division to identify slant asymptotes.

Key Rules: Analyzing $f(x) = \frac{p(x)}{q(x)}$

1. Factor First: Factor *both* numerator and denominator completely.

2. Hole vs VA: Factor in **num & denom** → **Hole** (plug cancelled value → y-coord).
Factor in **denom only** → **Vertical Asymptote**.

3. Horizontal Asymptote: $\deg(p) < \deg(q) \to y = 0 \mid \deg(p) = \deg(q) \to y = \frac{\text{lead}_p}{\text{lead}_q} \mid \deg(p) > \deg(q) \to \text{No HA}$

4. Slant Asymptote: When deg(p) = deg(q) + 1. Use **polynomial long division**; quotient = slant.

5. Direction near VA: Use a sign chart or test values just left/right of the VA to determine $+\infty$ or $-\infty$.

🔔 Mr. Kenny's Key Point

Always **factor first**. Same factor in num + denom = **hole**. Factor only in denom = **vertical asymptote**.

Compare **degrees** for horizontal: less → $y = 0$, equal → ratio of leading coefficients, more → slant (if difference = 1).

Worked Examples

Example 1 — Hole, VA & Horizontal Asymptote

Given: $f(x) = \dfrac{-4x - 16}{x^2 + 6x + 8}$

Step 1 — Factor: $\dfrac{-4x-16}{x^2+6x+8} = \dfrac{-4(x+4)}{(x+2)(x+4)}$

Step 2 — Hole: $(x+4)$ appears in both → **Hole at** $x = -4$.

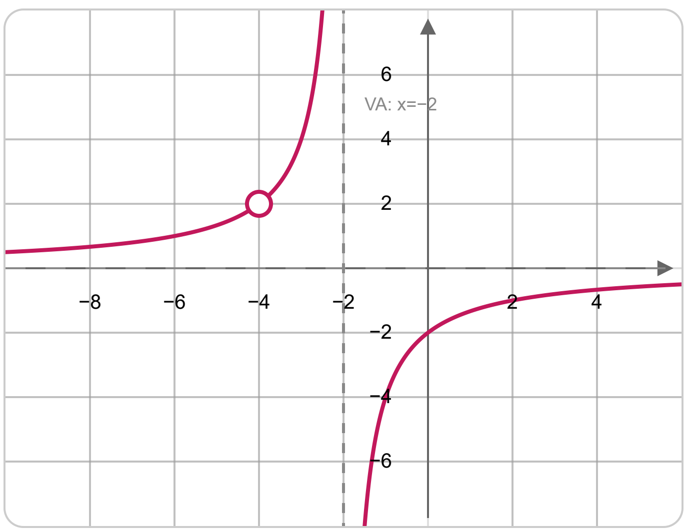
Reduced function: $\dfrac{-4}{x+2}$. Plug in $x = -4$: $\dfrac{-4}{-4+2} = \dfrac{-4}{-2} = 2$ → **Hole at** $(-4,\ 2)$

Step 3 — VA: Remaining denominator: $(x+2) = 0$ → **VA:** $x = -2$

Step 4 — HA: $\deg(\text{num}) = 1 < \deg(\text{denom}) = 2$ → **HA:** $y = 0$

Step 5 — Direction near $x = -2$: As $x \to -2^-$: $(x+2) \to 0^-$, numerator $\approx -4 < 0$ → $\dfrac{-4}{0^-} \to +\infty$

As $x \to -2^+$: $(x+2) \to 0^+$ → $\dfrac{-4}{0^+} \to -\infty$



$f(x) = \frac{-4x-16}{x^2+6x+8}$ — VA: $x = -2$, HA: $y = 0$, Hole: $(-4, 2)$

Example 2 — Two VAs, Hole & Horizontal Asymptote

Given: $f(x) = \dfrac{-3x+3}{x^3 - 4x^2 + 3x}$

Step 1 — Factor: Numerator: $-3(x-1)$.

Denominator: $x(x^2 - 4x + 3) = x(x-1)(x-3)$.

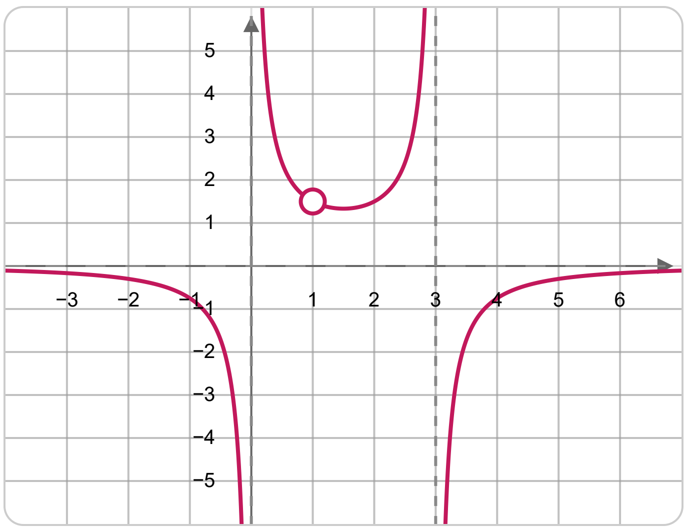
So: $\dfrac{-3(x-1)}{x(x-1)(x-3)}$

Step 2 — Hole: $(x-1)$ cancels → **Hole at** $x = 1$.

Reduced: $\dfrac{-3}{x(x-3)}$. Plug $x = 1$: $\dfrac{-3}{1 \cdot (1-3)} = \dfrac{-3}{-2} = \dfrac{3}{2}$ → **Hole at** $\left(1, \dfrac{3}{2}\right)$

Step 3 — VAs: Remaining denom zeros: $x = 0$ and $x = 3$ → **VA:** $x = 0$ **and** $x = 3$

Step 4 — HA: $\deg(\text{num}) = 1 < \deg(\text{denom}) = 3$ → **HA:** $y = 0$



$f(x) = \frac{-3x+3}{x^3-4x^2+3x}$ — VAs: $x = 0,\ x = 3$, HA: $y = 0$, Hole: $\left(1, \frac{3}{2}\right)$

Example 3 — Slant Asymptote (Polynomial Long Division)

Given: $f(x) = \dfrac{x^2 - x - 2}{x - 3}$

Step 1 — Factor & check for holes: Numerator: $(x-2)(x+1)$. Denominator: $(x-3)$. No common factors → **No holes**.

Step 2 — VA: $x - 3 = 0 \rightarrow$ **VA:** $x = 3$

Step 3 — Slant Asymptote: $\deg(\text{num}) = 2 = \deg(\text{denom}) + 1 \rightarrow$ **Slant asymptote exists!**

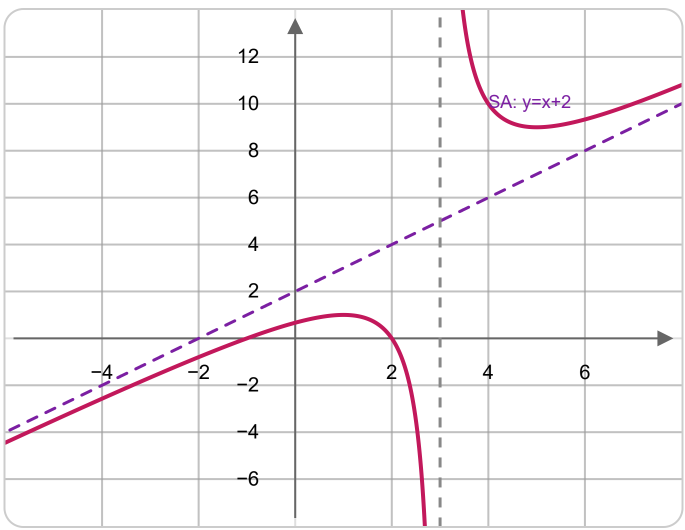
Polynomial long division: $(x^2 - x - 2) \div (x - 3)$

$$\begin{array}{r l}
 & x + 2 \\ \cline{2-2}
x - 3 \,) & x^2 - x - 2 \\
 & \underline{-(x^2 - 3x)} \\
 & \quad 2x - 2 \\
 & \quad \underline{-(2x - 6)} \\
 & \qquad 4 \quad (\text{remainder})
\end{array}$$

Quotient $= x + 2 \rightarrow$ **Slant Asymptote:** $y = x + 2$

Step 4 — Intercepts: x-intercepts: $(x-2)(x+1) = 0 \rightarrow x = 2$ and $x = -1$.

y-intercept: $f(0) = \dfrac{-2}{-3} = \dfrac{2}{3}$



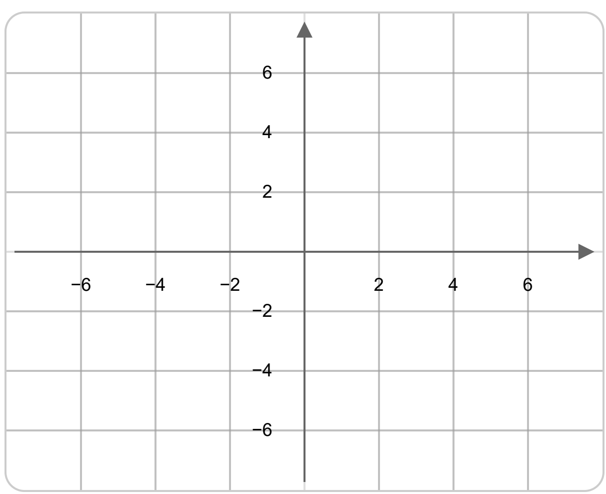
$f(x) = \frac{x^2-x-2}{x-3}$ — VA: $x = 3$, Slant: $y = x + 2$, No holes

Practice Problems

For each rational function: **(a)** *Factor completely and identify all holes (give exact coordinates).* **(b)** *Find all vertical asymptotes.* **(c)** *Find the horizontal or slant asymptote (show degree comparison or long division).* **(d)** *Sketch the curve on the grid using the asymptotes as guides.*

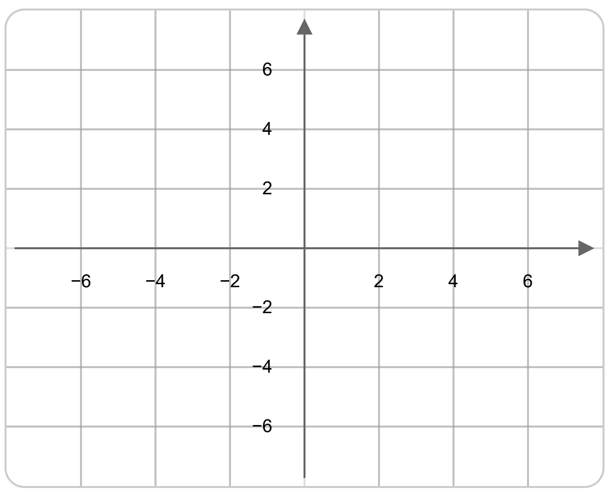
1. $f(x) = \dfrac{x-2}{x^2-4}$

Factored Form:	
Hole(s):	
VA:	
HA / SA:	
Direction Sketch:	



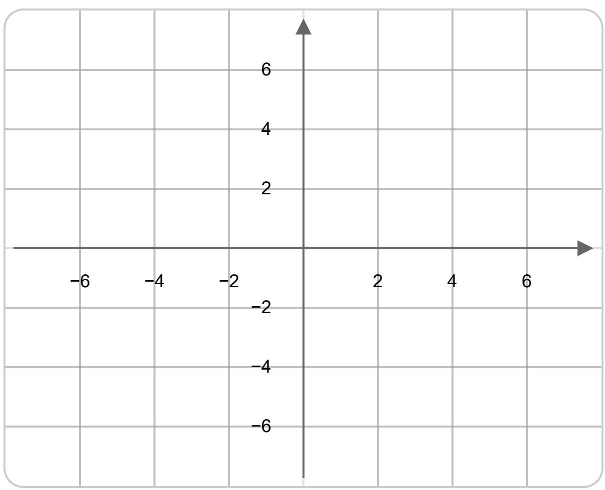
2. $f(x) = \dfrac{x^2 - 9}{x + 3}$

Factored Form:	
Hole(s):	
VA:	
HA / SA:	
Direction Sketch:	



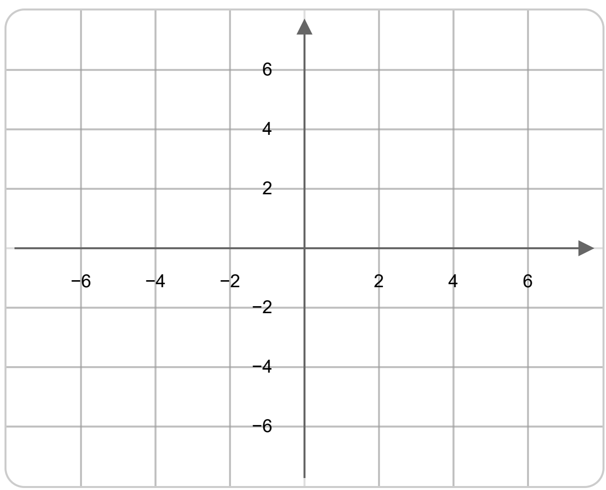
3. $f(x) = \dfrac{2x+6}{x^2+5x+6}$

Factored Form:	
Hole(s):	
VA:	
HA / SA:	
Direction Sketch:	



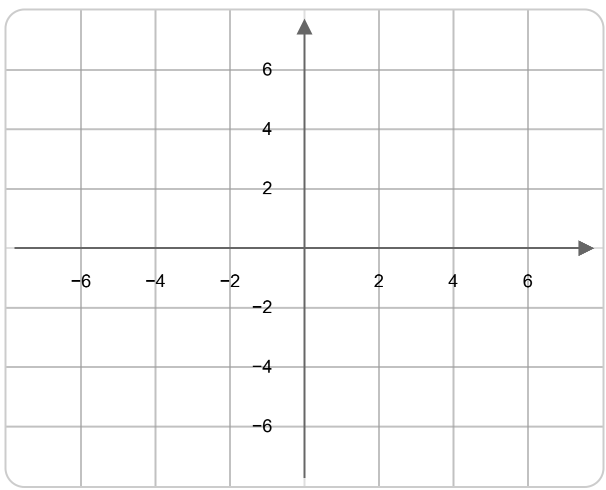
4. $f(x) = \dfrac{x^2 + x - 6}{-x^2 + 3x}$ ★ *Kenny's emphasis problem*

Factored Form:	
Hole(s):	
VA:	
HA / SA:	
Direction Sketch:	



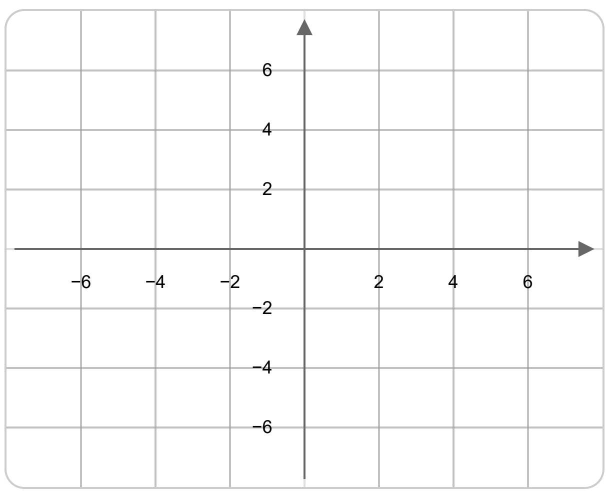
5. $f(x) = \dfrac{3x^2}{x^2 - 1}$

Factored Form:	
Hole(s):	
VA:	
HA / SA:	
Direction Sketch:	



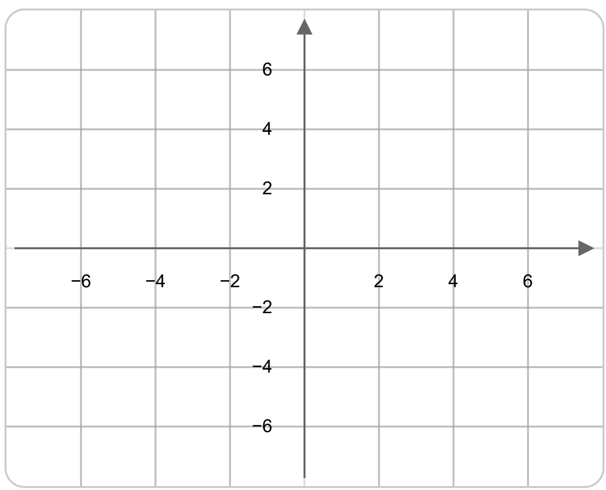
6. $f(x) = \dfrac{x^2 - 4}{x^2 - x - 6}$

Factored Form:	
Hole(s):	
VA:	
HA / SA:	
Direction Sketch:	



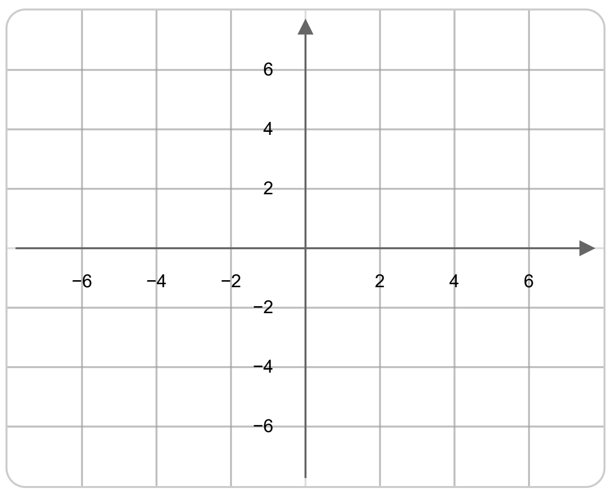
7. $f(x) = \dfrac{x^2 + 2x - 8}{x + 4}$ *(Hole & linear simplification)*

Factored Form:	
Hole(s):	
VA:	
HA / SA:	
Direction Sketch:	



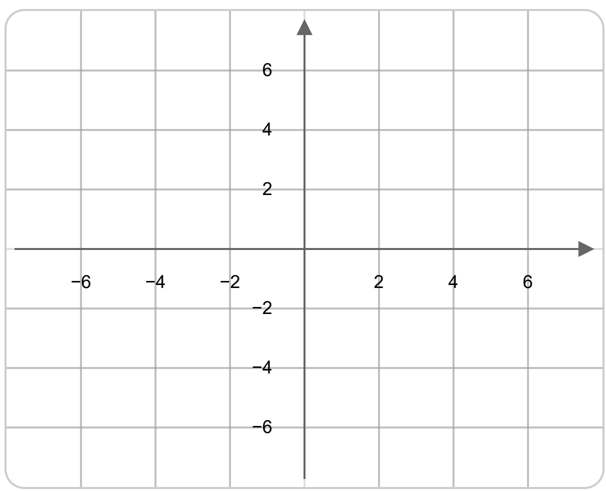
8. $f(x) = \dfrac{x^2 - 1}{x - 1}$ *(Hole & linear)*

Factored Form:	
Hole(s):	
VA:	
HA / SA:	
Direction Sketch:	



9. $f(x) = \dfrac{x^2 + 2x + 1}{x + 1}$ *(Perfect square — hole)*

Factored Form:	
Hole(s):	
VA:	
HA / SA:	
Direction Sketch:	



10. $f(x) = \dfrac{x^2 - 2x - 3}{x - 3}$ *(Hole & linear)*

Factored Form:	
Hole(s):	
VA:	
HA / SA:	
Direction Sketch: